

Advanced Computer Networks



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Transport Layer Basics



**DSSRG: Decentralized
Smart Systems Research
Group**

<https://sites.google.com/iitbbs.ac.in/dssrg>

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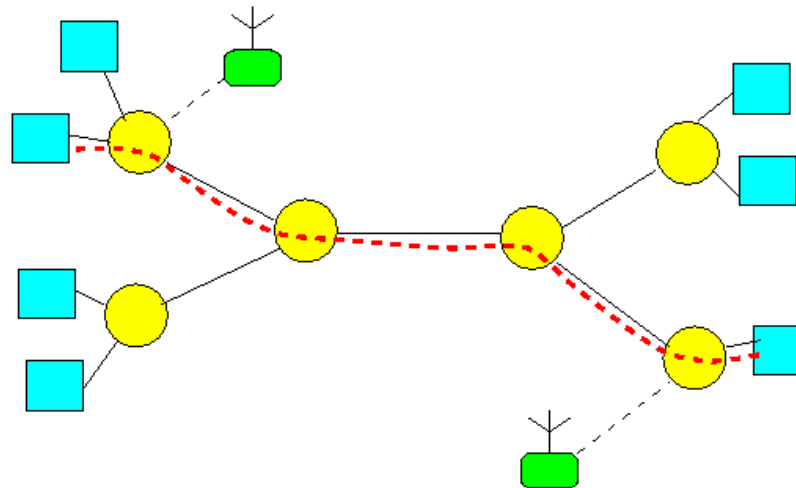
Introduction to the Transport Layer

- **Network** upto Network Layer ...

Network Layer: Transmits data (packets) between *any* pair of nodes

Datalink layer: Ensure *reliable* transfer of data between neighbor nodes

Physical layer: Transmits data (signals) between neighbor nodes



The Transport Layer

- To understand the services provided by the **Transport Layer**, you need to **keep** this in **mind**
- **Transport layer provide communication functions to user programs**
- **Transport layer** = the *first user* level layer in the **OSI architecture**
- The **Transport layer** provides **communication services** that a user (i.e.: *you*) can use



Different (user) programs have different needs:

- ***Most* program require *reliable* data communication**
- **E.g.: web browser, email, etc.**
- ***Some* programs can tolerate some data loss but need *speedy* packet delivery**
- **E.g.: audio chat tool**

Speed vs Reliability



Options ...

- There are *multiple* user programs running on a *single* computer !!!
- Catering to *different* needs - There are *multiple* Transport Layers available !!!
- The well-known Transport Protocols are:
 - **TCP (Transmission Control Protocol)** –
 - A heavy-weight (= lots of overhead) protocol that provide **reliable** transfer over the IP protocol
 - **UDP (User Datagram Protocol)** –
 - A light-weight (= minimal overhead) protocol that allow **users** to transmit **packets** as *quickly* as possible over the IP protocol



A lesser know Transport Protocol:

- **XTP (Xpress Transport Protocol):**
- A high speed protocol that was promoted by the XTP forum to replace the TCP protocol



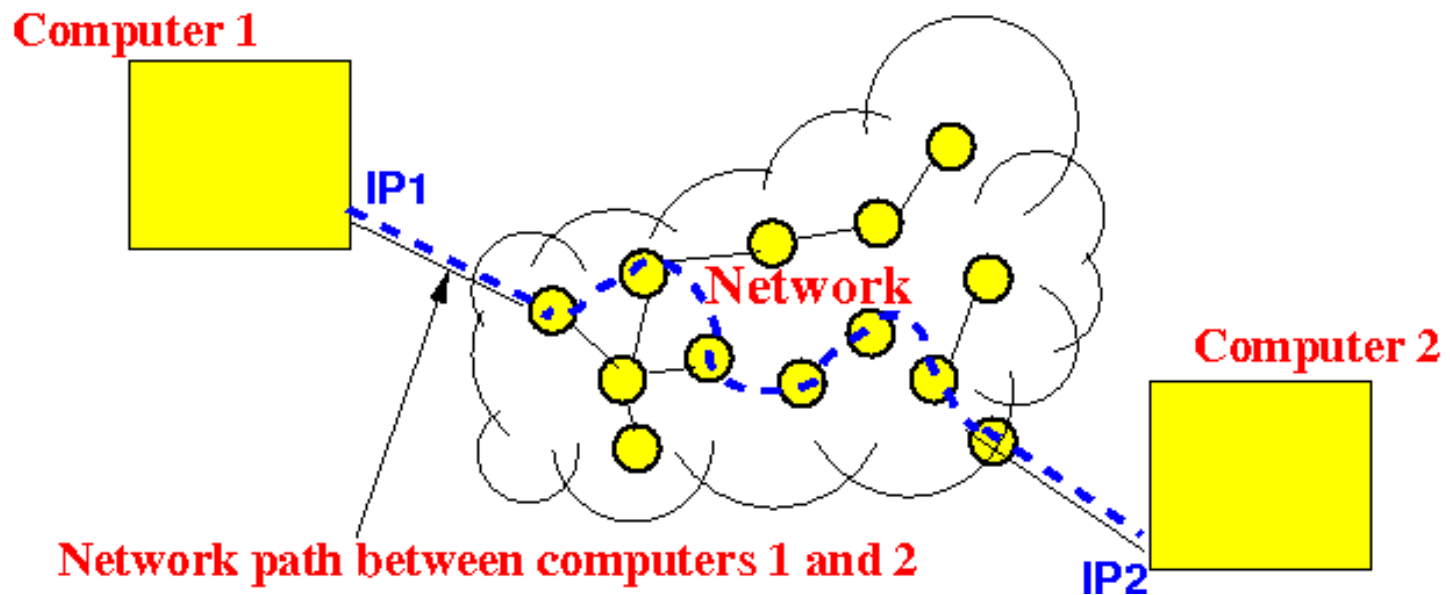
Commonly used functions provided by the Transport Layer

- **Reliable transfer:**
- The Transmission Control Protocol (TCP) provides *reliable* communication between 2 user application programs
- **Multiplexing:**
- *Every* transport protocol provide multiplexing

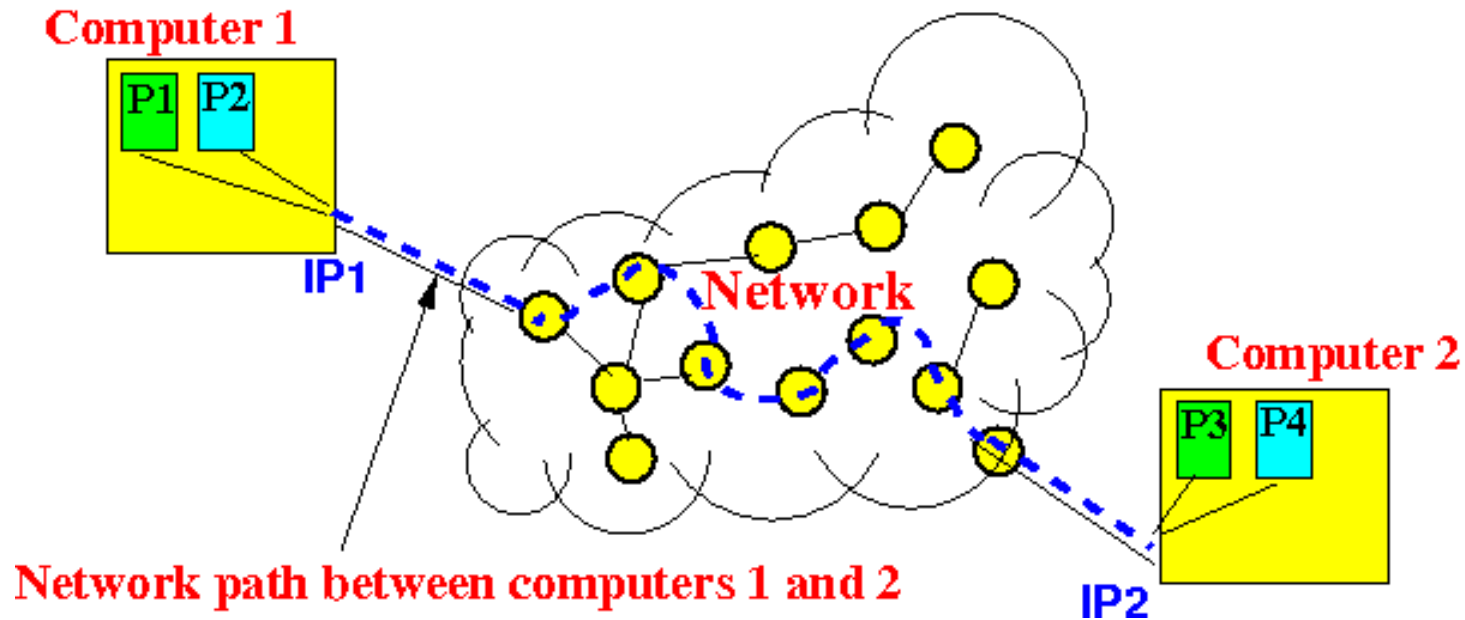


Why is multiplexing a necessary function in the Transport layer

- The **network layer** establishes a **communication path** between **two computers**
- **Packets** transmitted on this **communication path** are **identified** by:
 - The **source IP address**
 - The **destination IP address**



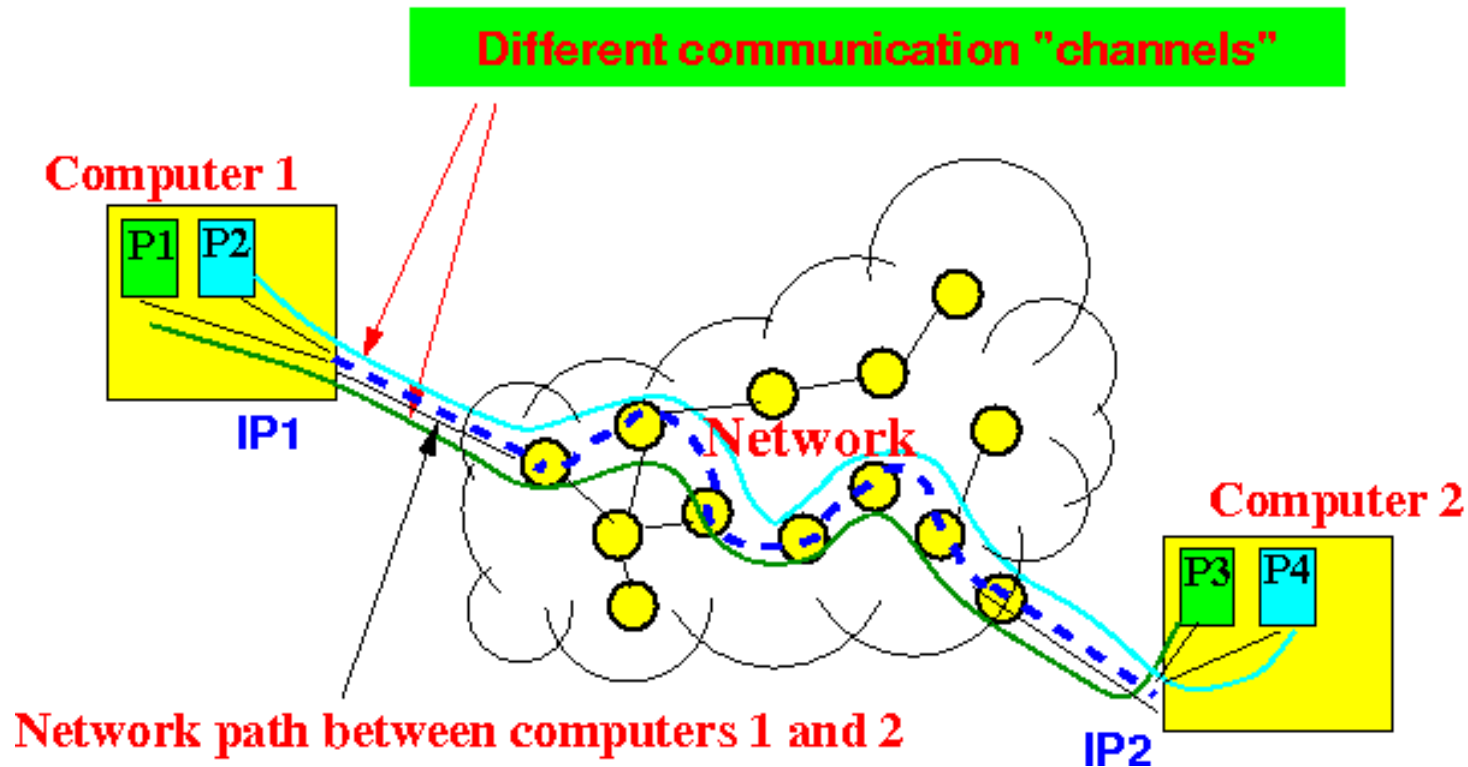
There are *multiple* application programs running on a *single* computer:



Different user application programs must be able to communicate with each other

Therefore

- The Transport layer must provide *different* communication "channels" *inside* the (single) network path to *each* pair of applications:



We call this *multiplexing* because:

- The **different applications** are *sharing* the *single* network connection established by the **Network Layer**....

Other important functions in the **Transport layer** deals with *efficient* operation:



Two other major tasks of Transport Layer

Flow control:

- Ensure that the sender's transmission speed does *not* exceed the receive capability of the receiver
- We make sure that the receiver buffers are *not* depleted...
- [Concerns buffer of the receiver]

Congestion control:

- The congestion control mechanism will adjust the transmission rate (of the sender) according to the current state of the network
- The goals of congestion control: → Transmit as *fast* as possible While *not* causing *congestion* in the network
- [Concerns buffer of the intermediate routers]



Two issues -

Overflowing the receivers buffer
(Flow control)

Vs

Congesting the network
(Congestion control)

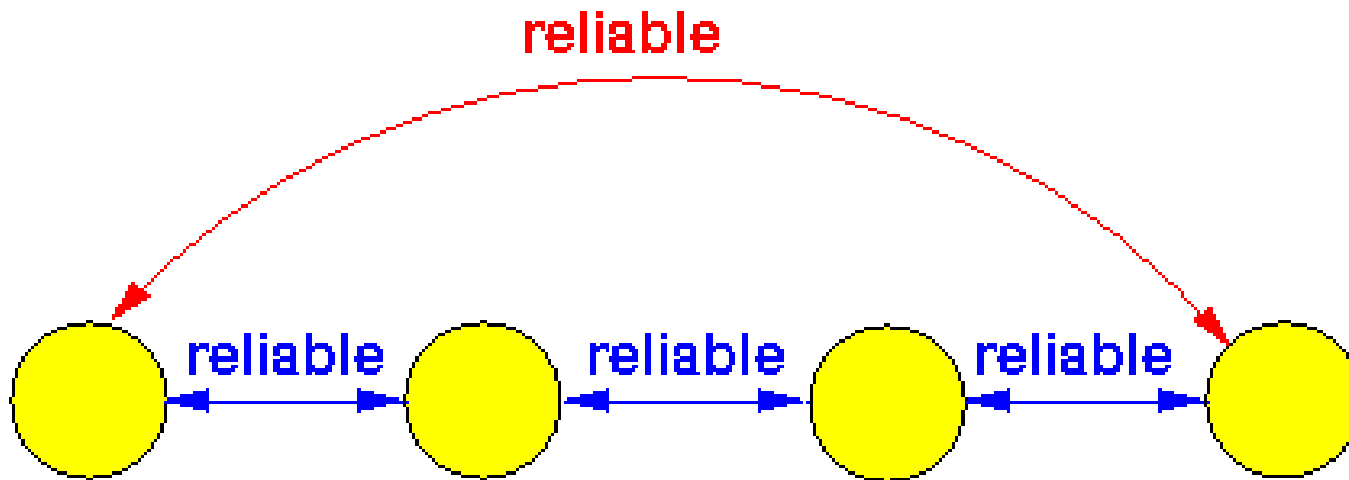


HOP-to-HOP vs END-to-END

- **Difference between hop-to-hop reliability (Datalink layer) and end-to-end reliability (Transport layer)**
- **A common misconception is:**
- **If messages are received *reliably* on a hop-to-hop basis:**
-



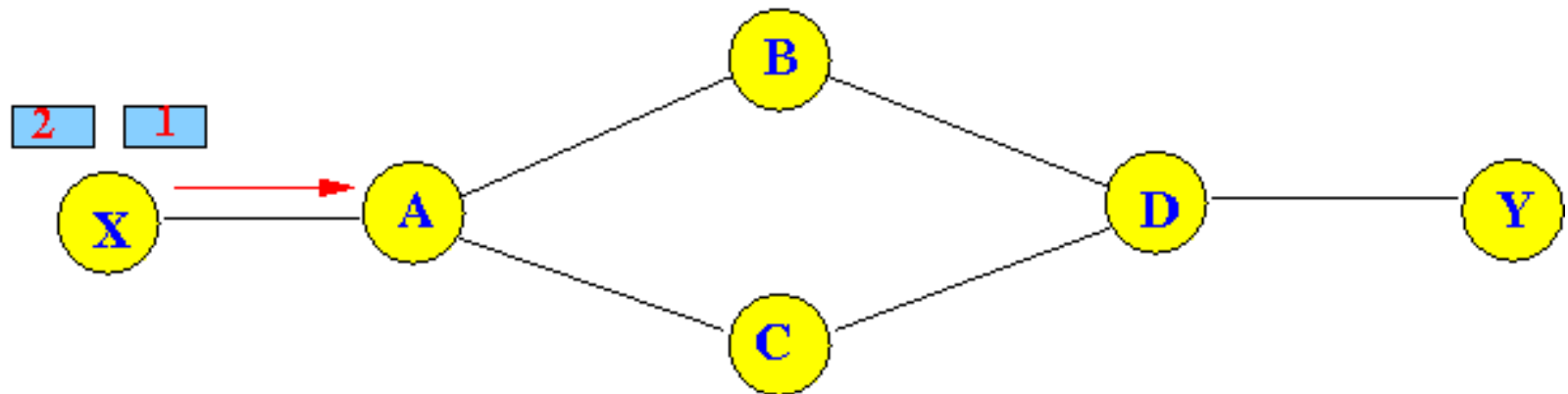
- then: the **messages** will be **received *reliably*** on an **end-to-end** basis



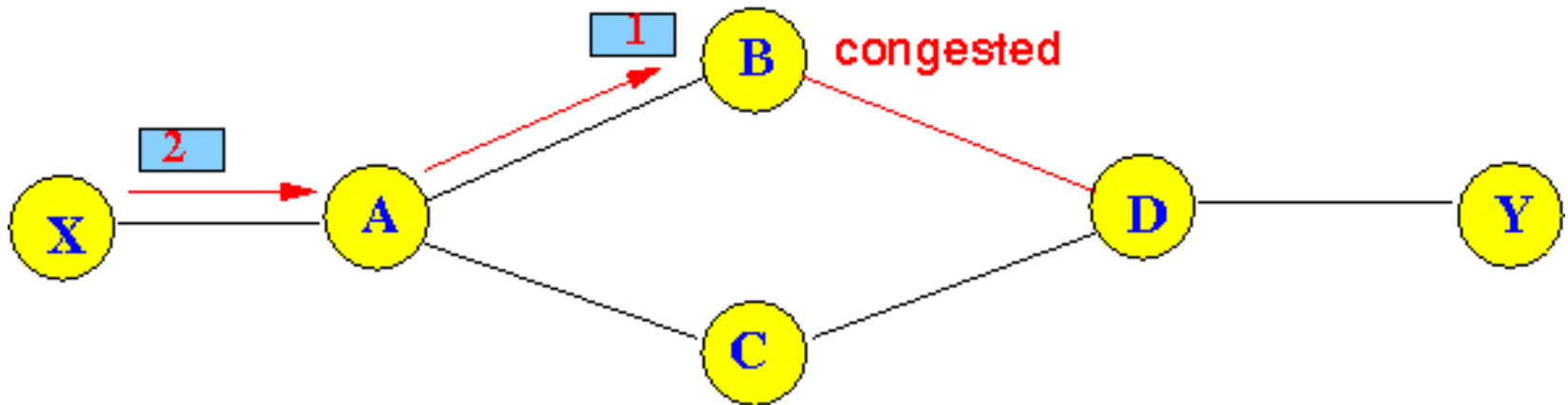
- The **above claim is *false* !!!**

What can cause end-to-end errors...

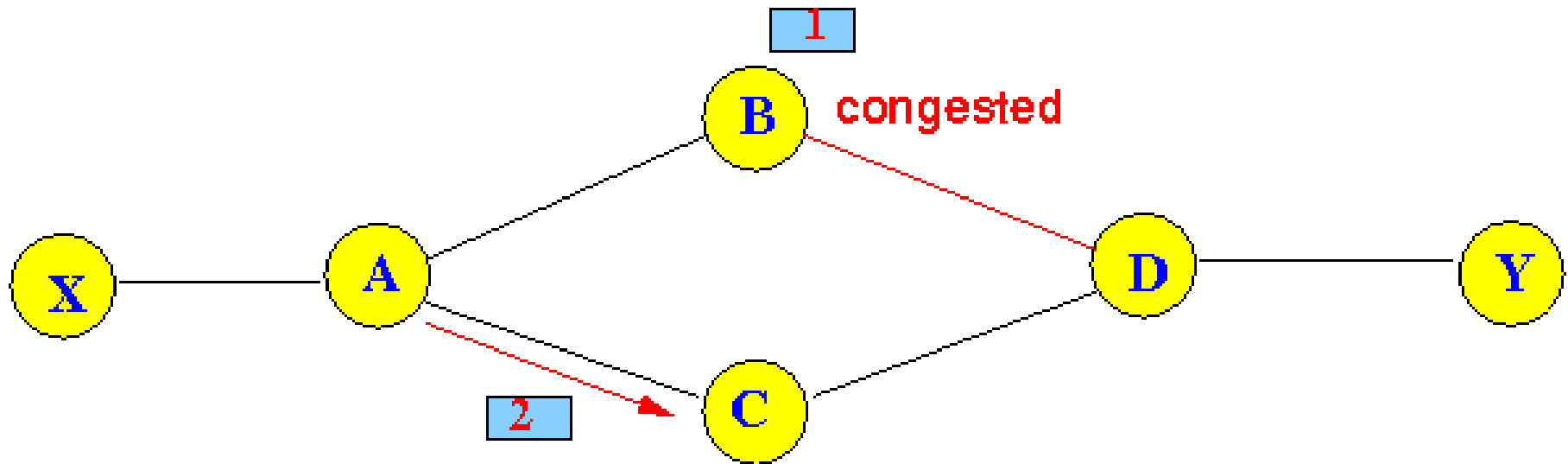
- In end-to-end communication, packets can be received *out-of-order*:
- Source X transmits 2 messages:



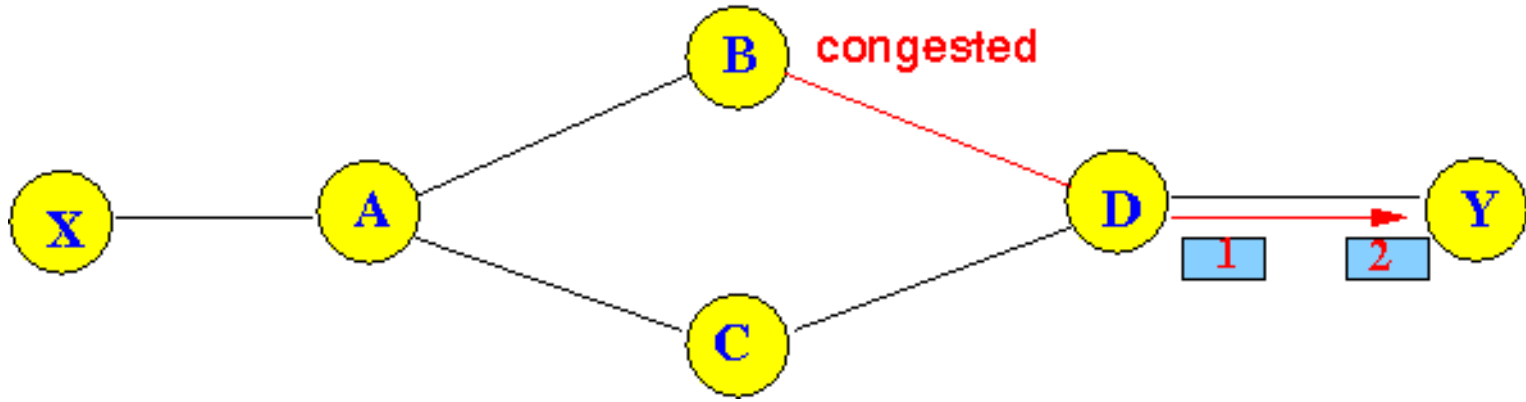
- **Packet 1** is routed into a **congested node B**:



- **Router A** *recomputes* routing tables and **route packet 2** to a *less congested* node:

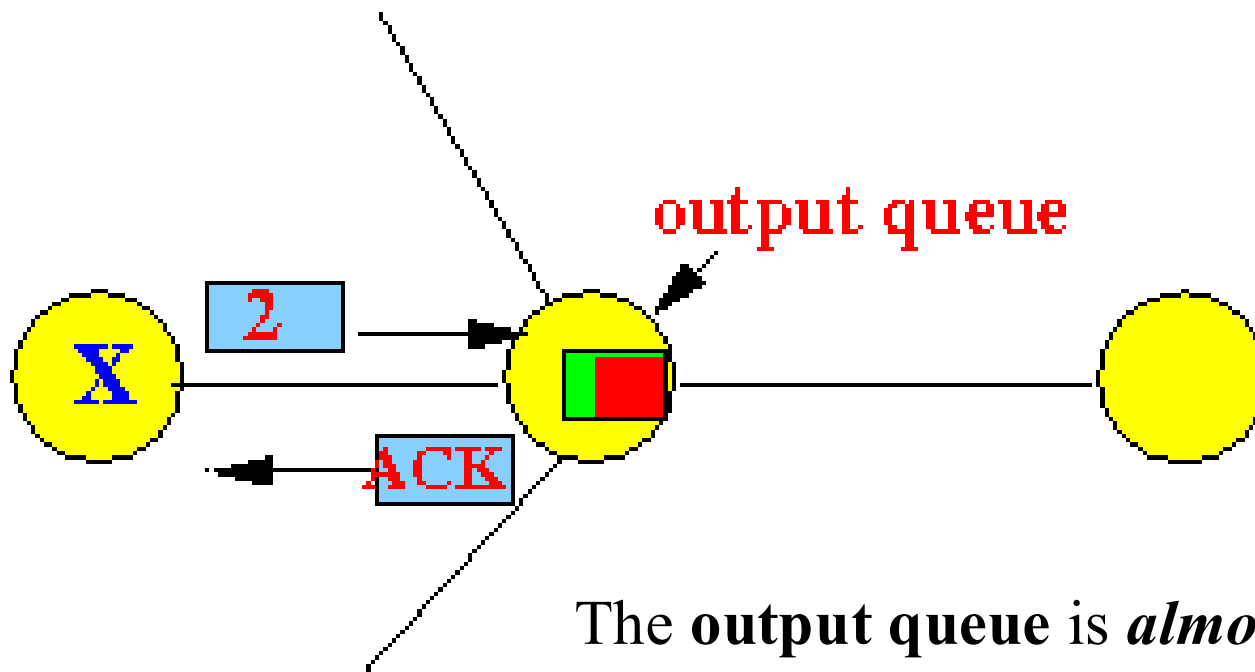


The packets will arrive *out-of-order*:

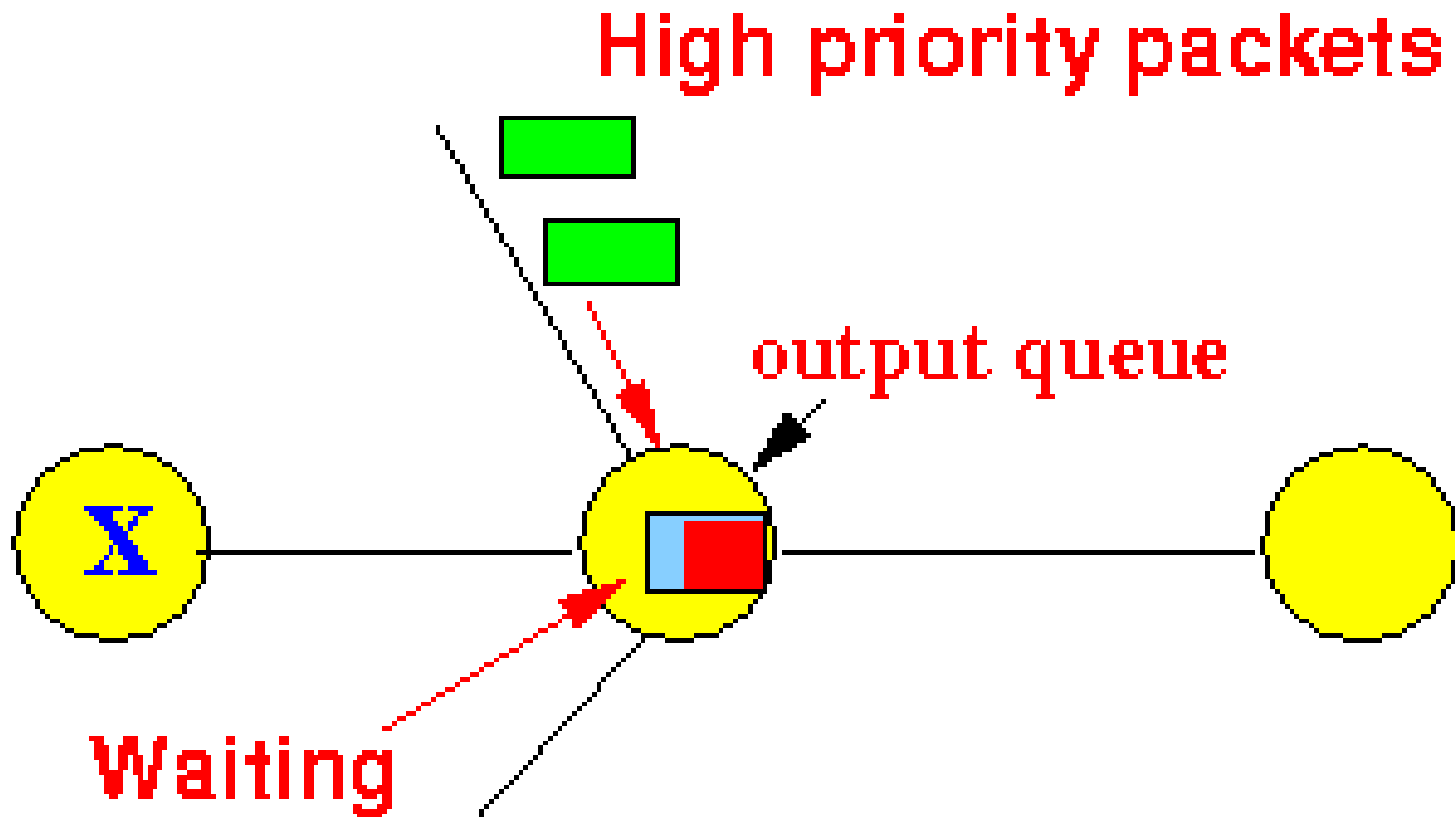


Case 2

- ***Received*** (and **acknowledged**) packets can be **deleted *later*** (to make space for a ***higher priority*** packet) !!!
- A router received (and **acknowledged**) a packet from X:



When some *high* priority traffic is received:



- Some **received** and **acknowledged** packet(s) will have to be *discarded* !!!

Thus...

- **Hop-to-hop reliability does *not* ensure end-to-end reliability**
- Therefore Transport Layer Needs to ensure Reliability separately...

